

Physical Fitness and Somatotype Profile of Male JCT Phagwara Football and Hockey Players

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ABSTRACT

The purpose of the study was to compare physical fitness and somatotype parameters of Hockey and Football players. The subjects for this study were 31 Football and 26 Hockey players of JCT Phagwara. Physical fitness tests like standing broad jump, 3 consecutive jumps, sit-ups, flex arm hang, 10x4 shuttle run, 30 m dash, 50 m dash, 2.4 km run were conducted to know about physical fitness level and antropometric procedures were used for the assessment of somatotype and body fat of selected groups. 't' test was applied to find out significance differences between fitness and antropometric values of Football and Hockey groups.

Footballers were found to be significantly better in leg strength, sprinting speed and abdominal strength than Hockey players; whereas, the Hockey players showed significantly higher level of arm and shoulder strength. A non-significant difference was found between Football and Hockey groups in agility level. A non-significant difference also was found in somatotype and percent body fat characteristics between Football and Hockey groups.

INTRODUCTION

JCT Football team was formed in the year 1971 by JCT limited, a leading textile mill of India. JCT Football team is one of the top teams in India and is placed 3rd in the I-League (2007-08); and was runners-up of the National Football League of the session 2006-07. JCT won Durand Cup 5 times in 1976, 1983, 1987, 1992 and 1996 and was runner-up 6 times during the year 1974, 1975, 1977, 1981, 1985 and 2006. JCT Football won Federation Cup 2 times in 1995 and 1996 (Source JCT Website).

So, it's important to know the physical fitness and somatotype profile of JCT players.

The United State President's Council on Physical Fitness and Sports defined the term physical fitness as, 'the ability to carry out daily tasks with vigor and alertness without undergoing fatigue, with ample energy to enjoy leisure time pursuits, and to meet unforeseen emergencies' General fitness implies the ability of a person to live most effectively with his/her potential, which depend upon the physical, mental,

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emotional, social and spiritual components of fitness which are highly interrelated. The primary components identified on physical fitness and sports were muscular strength, muscular endurance, and cardio respiratory endurance. However, latter on, Council also included some other motor performance components viz. agility, speed, flexibility and balance in physical fitness.

A study on physical fitness between Basketball and Football players has been reported by Nara, K. (2017). He found that Basketball players were significantly slower in 50 yard dash, better in standing broad jump, slower in shuttle run and 600 yard run than Football players. C.P. Singh (2015) published a correlation study on components of motor fitness with Football performance among Football players of Rajasthan. He found physical fitness component correlation with Football performance (800m run 0.87, Standing broad jump 0.28, Sit-ups 0.41, 10×4m shuttle run 0.89, 50m dash 0.93). Kaur et al (2001) reported a study on anthropometric and fitness profile of Asian Gold Medalist Kabaddi players. Studies on physical fitness parameters are limited in number. More research work required on physical fitness profile of athletes.

The concept and word of 'somatotype' was introduced by Sheldon (1940) in his book *The Varieties of Human Physique*. He defined somatotype as 'a quantification of three primary components determining the morphological

structure of an individual expressed as a series of three numerals, the first referred to endomorphy, the second to mesomorphy, and the third to ectomorphy.' The endomorphy refers to degree of fatness in proportion to height; whereas, mesomorphy musculo-skeletal development proportion to height. Ectomorphy refers to linearity i.e. height and weight proportion. Lot of somatotype studies have been conducted on players of different sports. Tanner (1964) published somatotype studies on Olympic athletes. Similarly, de Garay et al (1974) published anthropological and genetical studies on Mexico and Montreal Olympic players. Carter, JEL. (1984) conducted somatotype research work on Olympians. In India, Sodhi and Sidhu (1984) published a book *Physique and selection of sportsmen*. There are differences in somatotype components among players of same sport as well as players of different sports may also differ in somatotype components. But on an average Olympic top level male players have somatotype ratings 2-5-3. Thus, mesomorphic component is always dominating over endomorphic. In some sports like throwing event body weight is more on similar height as compared to runners, Basketballers and Volleyballers. The, body weight and fat also differs in different sports on similar height. Even in a same game, there are differences in body dimensions among players. Similarly, motor performance test also differs among players of same game as well as in different sports. So, study is very important to know

difference in fitness and somatotype profile among players of two games viz., JCT Football & Hockey. This study will provide differences in physical fitness and somatotype parameters among JCT Phagwara Football and Hockey players.

METHODOLOGY

The cross-sectional data were collected on 31 male Football players and 25 Hockey players of JCT Phagwara (Punjab) in May 1996. Anthropometric measurements were taken with standard instruments using techniques of Weiner & Lourie (1969). The anthropometric measurements were used for calculation of somatotypes and body fat. The somatotype components were assessed by applying computer equations of Carter (1980). Body density was calculated using skinfolds of biceps, triceps, subscapular and suprailiac sites by equations of Durnin and Womersley (1974). Body fat was estimated from body density by formulae of Brozek et al (1963.) For assessment of physical

fitness, the motor performance tests of standing broad jump, 3 consecutive jumps, 30m dash, 50m dash, shuttle run (10×4), sit-ups, flex arm hang and 2.4 km run time were conducted on players. Standard procedures were followed for taking these tests using recommended instruments. However, Hockey players did not undergo for 2.4 km run test.

Mean and standard deviation were computed, 'Student t' test was used to find out statistically significant differences between two groups (Football and Hockey players).

RESULTS & DISCUSSION

Table 1 shows the physical fitness tests of JCT Football and Hockey players. Standing broad jump in Football players is 2.44m and 2.28m in Hockey players. Standing broad jump used as a functional test to assess leg power. These findings indicates that former possess more explosive power of legs than latter. The value of 't' test also found to be significant. Football players also performed better in 3 consecutive

Table-1: Physical fitness tests of JCT Football and Hockey players.

Statistics	SBJ (m)	3CJ (m)	30m dash (sec)	50m dash (sec)	10×4 Shuttle run	Sit-ups (60 sec)	Flex Arm hang (sec)	2.4 km (sec)
Football players								
N	26	26	25	25	25	22	24	19
Mean	2.44	7.3	4.28	6.73	9.59	44	62.67	497.10
SD	0.11	0.68	0.21	0.27	0.46	11.7	14.98	28.46
Hockey players								
N	23	21	23	23	23	23	23	-
Mean	2.28	6.91	4.65	6.94	9.08	38.65	71.78	-
SD	0.13	0.41	0.19	0.24	1.22	3.59	16.80	-
't' value	4.62*	2.42*	6.41*	2.85*	1.88	2.05*	1.96*	-

Significant at 5% level

jumps (7.3m) than Hockey players (6.91m). Three consecutive jumps indicate horizontal and vertical power of the legs with a component of balance and coordination. In statistical terms former jumps significantly more than latter. Table 1 shows that Football players take significantly less time to complete 30m and 50m dash than Hockey players. 30m dash test is used for speed measurement, whereas; 50m dash measures running speed and acceleration. In agility test (shuttle run) both groups did not differ significantly. Flex arm hang test is used to measure upper body strength (muscular endurance of your elbow flexor and shoulder extensors). Hockey players perform better flex arm hang (71.78 sec) than Football players (62.69 sec). The comparison of flex arm hang test was significant between these groups (t 1.96). This indicates Hockey players have more upper body strength than Football players. Abdominal strength as measured by sit-ups is found to be significantly more (t 2.05) in Football players than Hockey players. The average time of Football players for 2.4 km run test was 497.10 sec.; however, Hockey players did not undergo for this test.

Table 2 shows the somatotypes and percent body fat of JCT Football and Hockey players. The mean somatotype of Football players was 3.71-4.47-2.74 whereas Hockey 3.43-4.5-2.71. The height weight ratio was 42.87 in Football and 42.75 in Hockey players. It is interesting to find here that the players of both the games don't differ from each other in somatotype components. Football players were slightly lesser leaner (fat 15.88%) than Hockey players (15.35%). In statistical terms body fat did not differ significantly in both groups. The findings of above results reveal that body structure and percent fat values were similar in JCT Football and Hockey players.

Table 3 shows the comparative studies of JCT Football players with Rajasthan Football players (C.P Singh, 2015), Haryana Football players (Kuldeep Nara, 2017), Haryana basketball players (Kuldeep Nara, 2017) and Indian Kabaddi players (Kaur R, et al 2001). JCT Football players perform significantly more standing broad jump than Rajasthan Football players, Haryana Football and basketball players. However, JCT players did not differ than Indian kabaddi players in standing broad jump. In case of sit-ups

Table-2 : Somatotype and percent body fat of JCT Football and Hockey players.

Somatotype	Football players (N= 31)		Hockey players (N= 25)		't' value
	Mean	SD	Mean	SD	
Endomorphy	3.71	1.31	3.43	1.25	0.81
Mesomorphy	4.47	0.88	4.50	0.78	0.13
Ectomorphy	2.74	1.0	2.71	0.62	0.13
HWR	42.87	1.39	42.75	0.84	0.38
Body fat (%)	15.88	3.92	15.35	3.71	0.51

Table-3: Comparative studies on physical performance profile with JCT Football players.

Statistics	Standing broad jump(m)	Sit-ups (60 sec)	10×4 Shuttle run	50m dash
JCT Football players				
N	26	22	25	25
Mean	2.44	44	9.59	6.73
SD	0.11	11.7	0.46	0.27
Rajasthan football players C.P Singh, 2015				
N	60	60	60	60
Mean	2.147	45.12	9.63	6.58
SD	0.845	6.35	1.43	0.89
't' value	2.63*	0.61	0.08	0.36
Haryana College football players by Kuldeep Nara 2017				
N	75	75	75	75
Mean	2.20	38.62	9.31	6.32
SD	0.11	3.23	0.36	0.52
't' value	2.58*	2.13*	2.77*	5.08*
Haryana College basketball players by Kuldeep Nara 2017				
N	75	75	75	75
Mean	2.31	38.18	11.13	6.90
SD	0.16	3.26	1.02	0.38
't' value	4.58*	2.31*	2.77*	2.44*
Asian Gold Medalist kabaddi players by Kaur R et al. 2001				
N	14			
Mean	2.46			
SD	0.168			
't' value	0.40			

Significant at 5% level

JCT players perform significantly better than Haryana Football and Basketball players; whereas, former did not differ than Rajasthan Football players. JCT Football players perform shuttle run test significantly slower than Haryana Football players but significantly faster than Haryana basketball players. However, they did not differ than

Rajasthan Football players in shuttle run timings. JCT Football players were better in 2.4km run test (endurance test) than Indian Gold medalist kabaddi players reported by Kaur R. et al, 2001 (497.10 sec SD 28.46). The former were significantly faster than latter in 2.4km run test ($t=6.62$).

Table 4 shows the comparative

Table-4: Comparative studies on physical performance profile with JCT Hockey players.

Statistics	Standing broad jump (m)	Sit-ups (60 sec)	10×4 Shuttle run	50m dash
JCT hockey players				
N	23	23	23	23
Mean	2.28	38.65	9.08	6.94
SD	0.13	3.59	1.22	0.24
Rajasthan football players C.P Singh, 2015				
N	60	60	60	60
Mean	2.147	45.12	9.63	6.58
SD	0.845	6.35	1.43	0.89
't' value	1.83	5.17*	0.83	0.88
Haryana College football players by Kuldeep Nara 2017				
N	75	75	75	75
Mean	2.20	38.62	9.31	6.32
SD	0.11	3.23	0.36	0.52
't' value	2.67*	0.04	0.89	7.93*
Haryana College basketball players by Kuldeep Nara 2017				
N	75	75	75	75
Mean	2.31	38.18	11.13	6.90
SD	0.16	3.26	1.02	0.38
't' value	0.91	0.56	7.31*	0.60
Asian Gold Medalist Kabaddi players by Kaur R et al. 2001				
N	14			
Mean	2.46			
SD	0.168			
't' value	0.26			

Significant at 5% level

studies of JCT Hockey players with Rajasthan Football players (C.P Singh, 2015), Haryana Football players (Kuldeep Nara, 2017), Haryana basketball players (Kuldeep Nara 2017) and Indian Kabaddi players (Kaur R. et al, 2001). JCT Hockey players perform significantly better than Haryana Football players in standing broad jump test. However, they did not differ than Rajasthan Football players, Haryana Basketball players and Indian Kabaddi

players in standing broad jump. In case of sit-ups JCT Hockey players perform significantly lesser than Rajasthan Football players. However, the former did not differ significantly than Haryana Football and Basketball players. JCT Hockey players were significantly faster than Haryana Basketball players in shuttle run test. However, former did not differ than Rajasthan Football and Haryana Football players in agility test. JCT Hockey players were significantly

Table-5: Other somatotype studies of Football players

Statistics	Somatotype	Study	Reference
Mean	3.71-4.47-2.74	JCT Football	Present study
SD	1.31-0.88-1.0		
Mean	2.33-4.63-2.90	Intervarsity East Zone west Bengal	Sukant Saha (2017)
SD	0.58-0.95-0.95		
Mean	1.5-3.44-3.39	College level Himachal Pradesh	Manohar lal (2015)
SD	0.71-1.04-1.13		
Mean	2.5-4.8-2.3	Professional Turkish Super League Soccer players (N 161)	Tahir Hazir (2010)
SD	0.79-0.93-0.73		
Mean	3.0-4.5-2.6	Professional Turkish First League Soccer players (N 144)	Tahir Hazir (2010)
SD	0.9-0.9-0.8		
Mean	1.7-5.6-2.6	Russian	Martirosov et al (1987)
Mean	2.0-5.5-2.0	South American	Rienzi et al (2000)
Mean	2.64-3.57-3.0	Low profile Lithuanian footballers	Boris et al (2015)
SD	1.04-0.47-1.05		
Mean	2.41-3.55-3.33	Elite Lithuanian footballers	Boris et al (2015)
SD	0.77-0.70-1.30		
Mean	4.5-6.5-1.5	Professional footballers National Football League	Wilmore et al 1976

slower than Haryana Football players in 50 m dash test. However, the former did not differ than Rajasthan Football and Haryana basketball players.

Comparative somatotype studies on Football players had been shown in Table 5. JCT Football players shows higher endomorphic (fatty component) than Intervarsity West Bengal (S.Saha, 2017), Himachal Pradesh (Manohar L., 2015), Professional Turkish Super and first league (T. Hazir, 2010), Russian (Martirosov et al, 1987), South

American (Rienzi et al, 2000), Low profile and high profile Lithuanian Footballers (Boris et al, 2015) but lesser than Professional National Football league (Wilmore et al., 1976). In case of mesomorphy, JCT Football players show lower values than Turkish super league, Russian, South Americans and professional national league. Ectomorphy (linearity i.e. height and weight proportion) has been found more in Himachal Pradesh Footballers, elite Lithuanian Football players; whereas,

Table-6: Other somatotype studies of Hockey players

Statistics	Somatotype	Study	Reference
JCT hockey players			
Mean	3.43-4.50-2.71	Present study	
SD	1.25-0.78-0.62		
Mean	2.3-4.6-2.7	Montreal Olympics	Carter et al, 1982
SD	0.8-0.9-0.8		
Mean	2.4-5.1-2.3	Argentina	Carter et al, 1981
SD	0.8-0.8-0.7		
Mean	1.9-4.5-2.7	New Zealand	Carter et al, 1981
	0.4-0.8-0.8		
Mean	2.7-4.0-3.2	Kenya-Malaysia	Carter et al, 1981
	1.0-0.6-0.8		
Mean	2.8-5.1-2.0	Nigeria	Toriola et al, 1985
	0.6-0.7-0.8		
Mean	2.4-5.4-2.6	South Australia	Withers et al, 1986
SD	0.6-0.8-0.6		
Mean	4.6-6.5-1.4	San Diego State University	Carter 1970
	1.3-1.0-0.8		
Mean	3.2-6.2-2.16	University of Iowa	Carter 1970
	1.1-0.6-0.7		
Mean	3.6-5.5-2.1	Oregeon College	Allen, 1965
	0.4-0.6-0.8		
Mean	4.2-5.4-1.9	University of Oklahome	Votto, 1976

lesser professional national Football league players than JTC Football players.

The mesomorphic and ectomorphic component of JCT Hockey players resembles with Montreal Olympic Hockey players (1976); but, their endomorphic component (degree of fatness) was higher than Olympian Hockey players. However, the endomorphic component of JCT players was lesser than San Diego State University, Oregon College and Oklahoma University Hockey players. JCT Hockey players were lesser mesomorphic than Nigerian, South Australian, San Diego State University, University of Iowa, Oregon College and University of Oklahoma Hockey players.

CONCLUSION

1. A significant difference exists between football and Hockey players in physical fitness parameters.
2. Football players were found to be stronger in leg strength, sprinting speed and abdominal strength
3. Hockey Group shows a significantly higher level of agility and arm and shoulder strength.
4. There is non-significant difference between Football and Hockey group in somatotype.
5. Body composition (percent body fat) was found to be similar in selected group.
6. On the basis of results we can conclude that selected groups have similar body structure and composition but different in physical fitness, It indicates that body structure alone may not lead to higher physical fitness level.

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